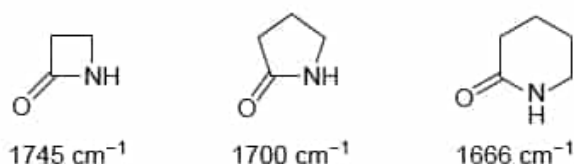


The infrared stretching frequencies for lactam carbonyl groups are shown below.



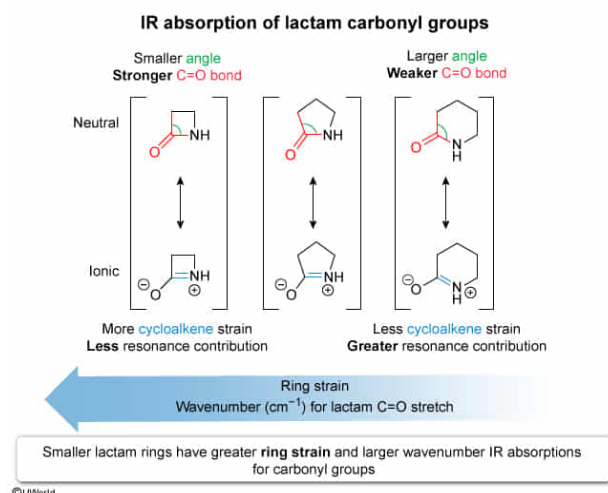
Which of the following best explains the difference in observed IR stretching frequencies?

- ☐ A. Molecular weight [9%]
☐ B. The inductive effect [14%]
☐ C. Acidity
✔ ☒ D. Ring strain [75%]

Correct

75% answered correctly

Explanation:



The properties of **lactams** (cyclic **amides**) result from the **delocalization** of electrons. Delocalization of the nitrogen lone pair into the carbonyl generates a resonance form with both charge separation and a different pattern of single and double bond character. The hybrid resonance structure of a lactam reflects the extent of electron delocalization.

Infrared (IR) spectroscopy is an analytical technique in which a sample is irradiated with IR light and a **spectrometer** detects and records the percentage of radiation that passes through the sample over a range of frequencies. Absorption of IR radiation causes **vibrational and rotational motion** in a molecule. Different bonds and functional groups absorb IR radiation at different **intensities** and different frequencies, and therefore the signal from each bonding environment appears in a characteristic

region of the IR spectrum.

The data shows that a smaller lactam ring size leads to a higher wavenumber carbonyl stretch in their IR spectrum. Ring strain includes both **angle strain** and **torsional strain** and is greater for smaller ring sizes. In a lactam, the partial double bond character of the C-N bond leads to angle strain when bond angles significantly deviate from the preferred 120° bond angle of a planar sp^2 carbon atom. To minimize ring strain, smaller lactam rings demonstrate *less* resonance delocalization and *less* C-N double bond character. Smaller lactam rings also have angle strain for the lactam carbon atom. Together, these features *strengthen* the C–O bond and lead to a higher wavenumber for the carbonyl stretching vibration.

Consequently, **ring strain** is responsible for the observed trend in IR stretching vibrations among lactams.

(Choice A) By **Hooke's Law**, atoms act as weights on either side of a spring (bond), where heavier atoms lead to slower frequencies of oscillation. The atoms contributing to the C–O bonds *do not vary* between the three molecules.

(Choice B) The **inductive effect** describes how electron density can be donated or removed through sigma bonds. The sigma bond environment around the three lactam functional groups is *unchanged*.

(Choice C) The N–H protons of a lactam are only weakly acidic and *do not vary* significantly between lactam molecules of varying ring size.

Educational objective:

The spectral properties of lactams (cyclic amides) can vary with ring size. In infrared spectroscopy, a lactam's carbonyl stretch appears at a higher wavenumber when the molecule has significant ring strain.

Time spent: 0 sec

Subject:

Organic Chemistry

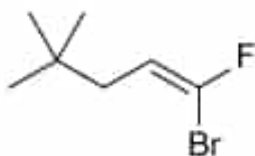
QID: 403959

System:

Functional Groups and Their Reactions

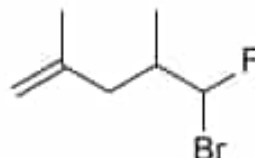
Which structure corresponds to (1Z)-1-bromo-1-fluoro-2,4-dimethylpent-1-ene?

☐ A.



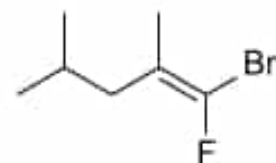
[4%]

☐ B.



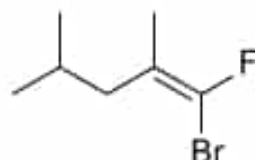
[2%]

☐ C.



[30%]

✓ ☒ D.

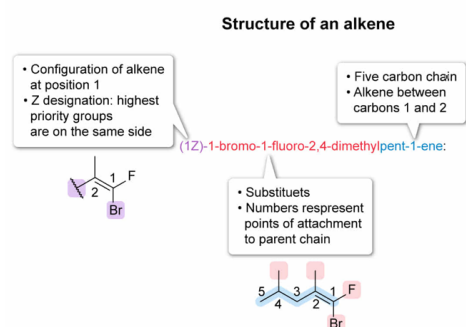


[63%]

Correct

62% answered correctly

Explanation:



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IUPAC provides a systematic method of naming organic compounds. A compound's **parent chain** is the longest continuous chain of carbon atoms that contains the greatest number of **highest-priority functional groups** and is numbered to provide the highest-priority functional groups with the lowest possible number. The parent chain name includes a **prefix** denoting the number of carbon atoms in the chain, a middle portion denoting the presence of select pi-bond functional groups, and a suffix

denoting the quantity and type of highest-priority functional group.

Lower-priority functional groups are named as **substituents**, which include a numerical position of attachment to the parent chain. Substituent chains are numbered from the point of attachment to the parent chain and named using a prefix to indicate the carbon chain length and a -yl suffix. Double bonds that are **stereocenters** are assigned either an **E- or Z-configuration** using the rules for **stereochemical priority**.

The *pent-* prefix indicates a five-carbon parent chain, whereas the suffix *-1-ene* indicates an alkene functional group between carbons 1 and 2. The preceding portions of the IUPAC name (1-bromo-1-fluoro-2,4-dimethyl) indicates that there are a total of four substituents:

- a bromine (carbon 1)
- a fluorine (carbon 1)
- two methyl groups (carbon 2 and carbon 4)

The (1Z) indicates that the double bond at carbon 1 has a Z configuration, where the highest-priority groups on each half of the C=C are on the *same side*. The bromine has higher priority than the fluorine due to its greater atomic number, whereas the rest of the pentane chain has higher priority than the 2-methyl group.

Choice D is the corresponding structure of (1Z)-1-bromo-1-fluoro-2,4-dimethylpent-1-ene.

(Choice A) This structure has two methyl groups at carbon 4, rather than one at carbon 2 and one at carbon 4.

(Choice B) This structure incorrectly places the double bond on the opposite side of the parent chain (ie, starting at carbon 4 instead of carbon 1).

(Choice C) This structure has the incorrect configuration for the alkene, where the highest-priority groups are on *opposite* sides of the double bond (*E* configuration).

Educational objective:

IUPAC nomenclature provides a systematic way of naming organic compounds. An IUPAC name can be used to determine the structure of the corresponding molecule.

Time spent:0 sec

Subject:

Organic Chemistry

QID:403960

System:

Functional Groups and Their Reactions